Derek Li

Toronto, Canada (Remote-friendly) · officials.yufengli@outlook.com
Portfolio: https://yufengli.vercel.app · Phone: 647-949-3090

Full-Stack Engineer (7 Years) — SaaS, Cloud-Native Architecture, Scripting & DevOps

SUMMARY

Full-stack engineer with seven years of experience delivering reliable, scalable SaaS products. Recognized for sound technical judgment, clear communication of trade-offs, and end-to-end execution from design through production. Since 2022, when GitHub Copilot was in technical preview, I've embraced AI coding assistants and integrated a range of AI-assisted development tools into my workflow to enhance productivity and code quality. I value ownership, cross-team collaboration, thoughtful engineering, clear architecture, and high standards for code quality, with a focus on simplifying systems and removing bottlenecks.

CORE SKILLS

Languages: TypeScript, JavaScript, Rust, Python, SQL

Frontend: React, Next.js, Angular, Tailwind CSS (Bootstrap → Tailwind refactors), Three.js

Data: PostgreSQL, MySQL, MongoDB, Redis, data migration scripting

Testing & Quality: Jest (unit/integration), Figma, test data tooling

Media/Automation: FFmpeg pipelines (MP4 → HLS/M3U8, audio+image composition), Puppeteer rendering

AI & Agents: Claude Code, MCP, Gemini (Vertex AI, Live API) tool calling, RAG

Backend: Node.js (Express/Nest patterns), Actix Web (Rust), GraphQL, REST, WebSockets

Cloud & DevOps: AWS (Cognito, CloudFront, ECS/ECR, S3, RDS, Redis, SES, OpenSearch), Terraform (IaC), Docker, GitHub Actions (CI/CD)

Other: 3D web graphics, background workers, performance optimization, Stripe billing (subscriptions, webhooks, proration/idempotency), message queues (BullMQ/Redis, SQS)

PROFESSIONAL EXPERIENCE

LearnFormula · Senior Full-Stack Engineer (Continuing Professional Development e-Learning) • Toronto, Canada — 2023 – Present

- Architected a production AI agent platform: Gemini tool-calling agents that research course topics, vet student requests, and draft and monitor email campaigns on a shared, budget-capped tool loop. Nothing sends without human approval, and guardrails live in code, not prompts.
- Made AI coding agents (Claude Code) my main way of building: 157 merged PRs in 2026, about 85% of my work since May. I write the design spec and plan, agents implement it task by task with TDD, each task gets an independent review, and multi-agent adversarial review runs before merge.
- Led the production migration from MySQL to PostgreSQL (148 tables, ~108M rows). Turned 162 app-level relations into 260 enforced foreign keys, verified against full production data, and cut over to

Terraform-managed Aurora PostgreSQL behind feature flags.

- Shipped core features of AIFormula, an AI learning assistant: a Gemini Live voice interviewer that turns an instructor's interview into a course, a multi-chat workspace with live course and credit cards, and course edits proposed as cards the learner applies.
- Architected a unified background processing platform (Redis + BullMQ), migrating legacy MySQL + PHP cron jobs; reduced campaign scheduling from ~1 hour to ~10 seconds, improved reliability, and enabled shared job logic across Node.js services.
- Rebuilt the Stripe integration around one idempotent sync layer with event-time ordering and nightly reconciliation with drift alerts, hardened through two rounds of multi-agent adversarial code review.
- Co-led the migration from AWS Elastic Beanstalk to Terraform-managed infrastructure (ECS, S3, RDS, Cognito, OpenSearch, CloudFront), codifying services for reproducible, reviewable environments. Reviewed and optimized AWS architecture (rightsizing, service selection, caching), reducing operational cloud spend by ~20%.
- Delivered an in-house FFmpeg media pipeline replacing AWS MediaConvert (MP4 → HLS, audio + image composition), integrated with the queue and cutting transcoding/streaming costs by ~50%.
- Led prototyping of an AI-assisted content pipeline from source readings to structured content, transcripts, TTS voiceovers, auto-generated slides, and rendered video.
- In collaboration with app developers, oversaw adoption of AWS Cognito to unify identity across web and React Native, simplifying SSO domain configuration and token strategy.

> *Tech: React, Next.js, Node.js, TypeScript, Redis, BullMQ, Stripe, AWS (Cognito, SES, CloudFront, ECS, S3, RDS, OpenSearch), Mailgun, Sendgrid, Terraform, Docker, FFmpeg, Puppeteer*

Spaice · Full-Stack Developer (Interior Design SaaS: 3D Editor, Quotes & Orders) · Toronto, Canada **2019 – 2022**

- Delivered a browser-based 3D editor with Three.js (materials, lighting, transforms), enabling interactive interior visualization and client-ready proposals.
- Participated as a core member and developed Rust-based GraphQL services for quoting, orders, and project data, evolving schemas to support product expansion.
- Established Amazon SQS for asynchronous task decoupling and architected gRPC service-to-service communication to standardize contracts and reduce coupling.
- Developed Stripe subscriptions/checkout/invoicing integration and established secure, idempotent webhook processing to maintain billing integrity.
- Delivered customer/admin web UIs (React/Next.js + GraphQL) for order management, client project tracking, and billing visibility.
- Managed CI/CD via GitHub Actions (build/test/deploy pipelines), reducing manual deploy friction and improving release cadence.
- Researched and presented Sanity.io CMS integration, empowering non-technical teams to publish marketing and catalog content independently.

> *Tech: Three.js, React/Next.js, Rust (Actix Web), GraphQL, Stripe, Amazon SQS, gRPC, Sanity.io, GitHub Actions, Docker, AWS*

EDUCATION

- University of Toronto — Honours Bachelor of Science in Computer Science (2014 – 2019)